***Jailbreaking attacks*** are algorithms that search for **input prompts** that cause a **targeted LLM** to generate **objectionable content**.

# What is a *jailbreaking attack*?

$$\mathrm{JB}(R) = \mathrm{JB}(R, G) = \begin{cases} 1 & R \text{ is objectionable} \\ 0 & \text{otherwise} \end{cases}$$

(e.g., checking for a particular string, LLM-as-a-judge, Llama Guard classifier, etc.)

Sources: (**GCG**; Zou et al., 2023), (**LLM-as-a-Judge**; Zheng et al., 2023), (**Llama Guard**; Inan et al., 2023).

$$\max_{P} \quad \Pr\left[\mathrm{JB}\Big(\mathrm{LLM}(P), \boldsymbol{G}\Big) = 1\right]$$

| | | |
|---|---|---|
| **input prompts** | **targeted LLM** | **objectionable content** |
| contain | is expressed by | evaluated by |
| goal string $G$ | $P \mapsto \text{LLM}(P) =: R$ | judge function $\text{JB}(R)$ |

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